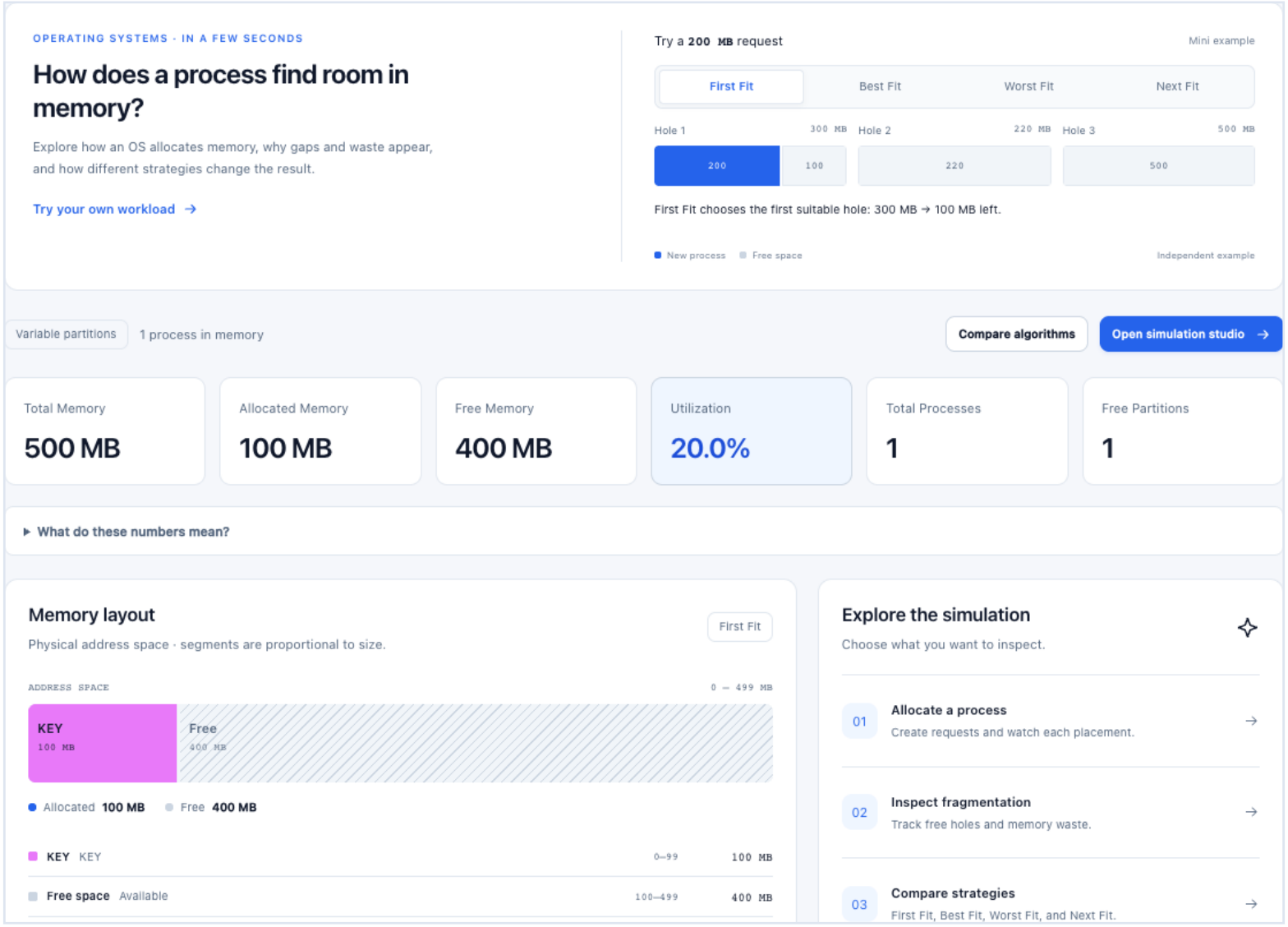
Website overview

Operating Systems project | Umang Vekariya | Sarthak Dhawale | Aditya Khunt

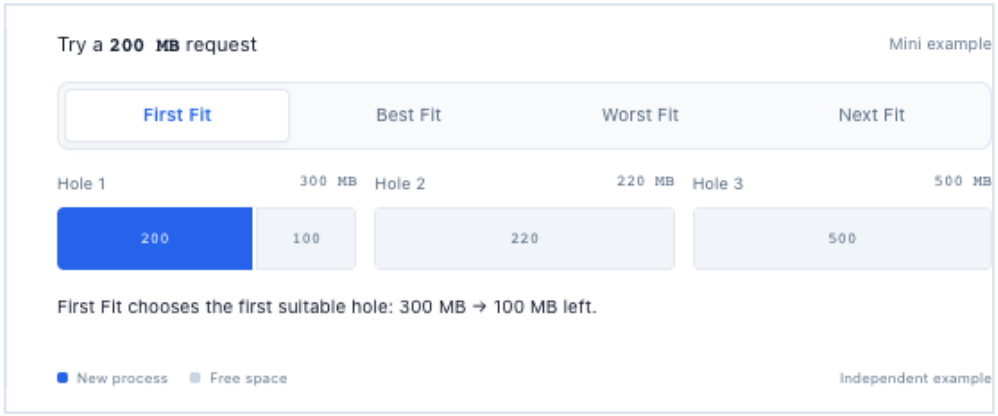


Simulate contiguous memory allocation, inspect fragmentation, and compare four placement strategies.

Try the four allocation strategies

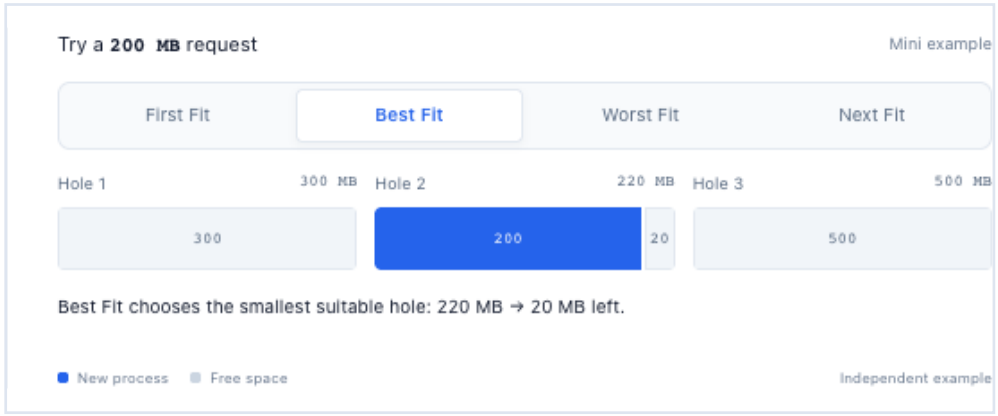
The Overview mini example places a 200 MB request into one of three free holes.

First Fit



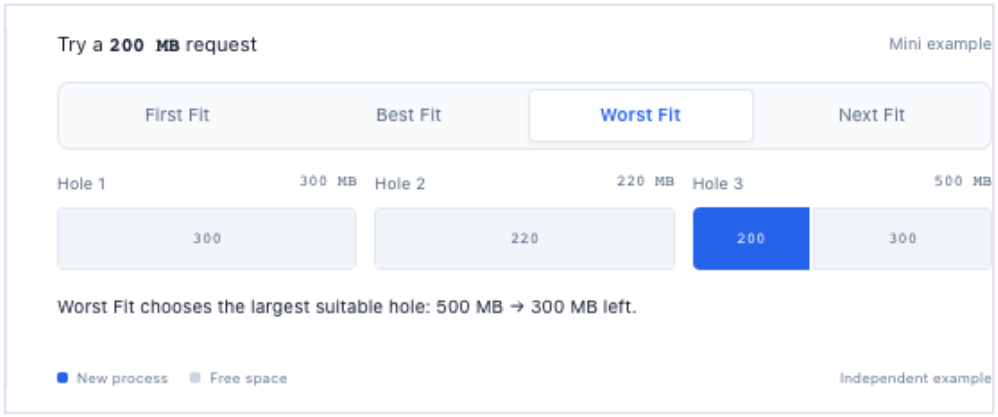
300 MB hole -> 100 MB left

Best Fit



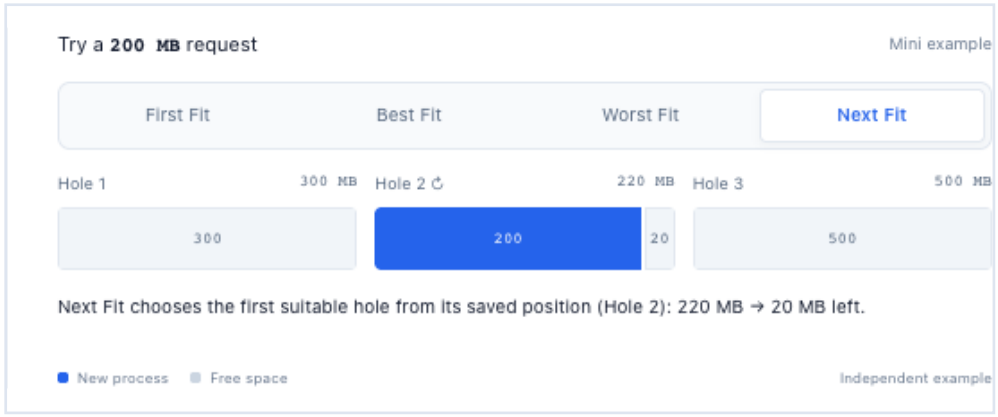
220 MB hole -> 20 MB left

Worst Fit



500 MB hole -> 300 MB left

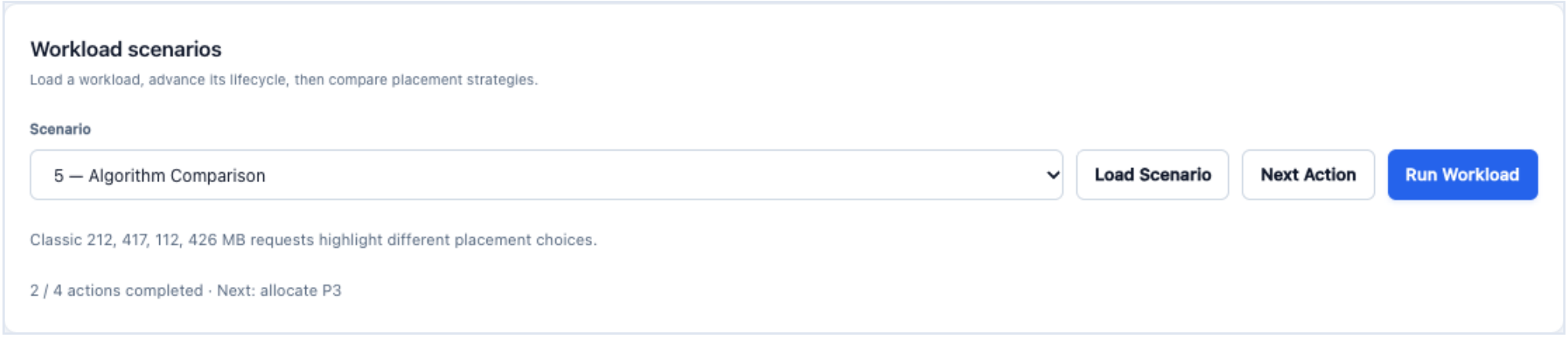
Next Fit



Resume at 300 MB -> choose the 220 MB hole

Load and run realistic workloads

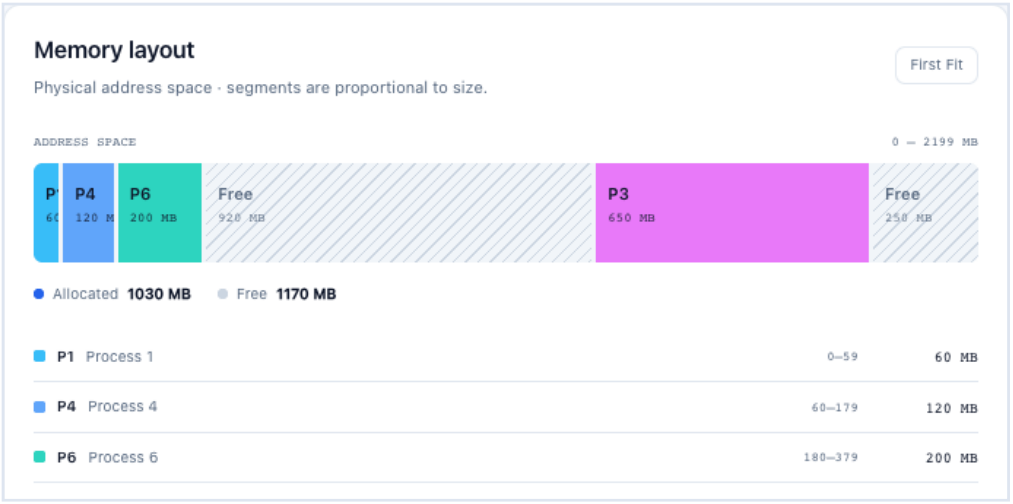
Normal, fragmentation, large, mixed, and classic comparison workloads are built into the website.



Normal workload - First Fit / variable



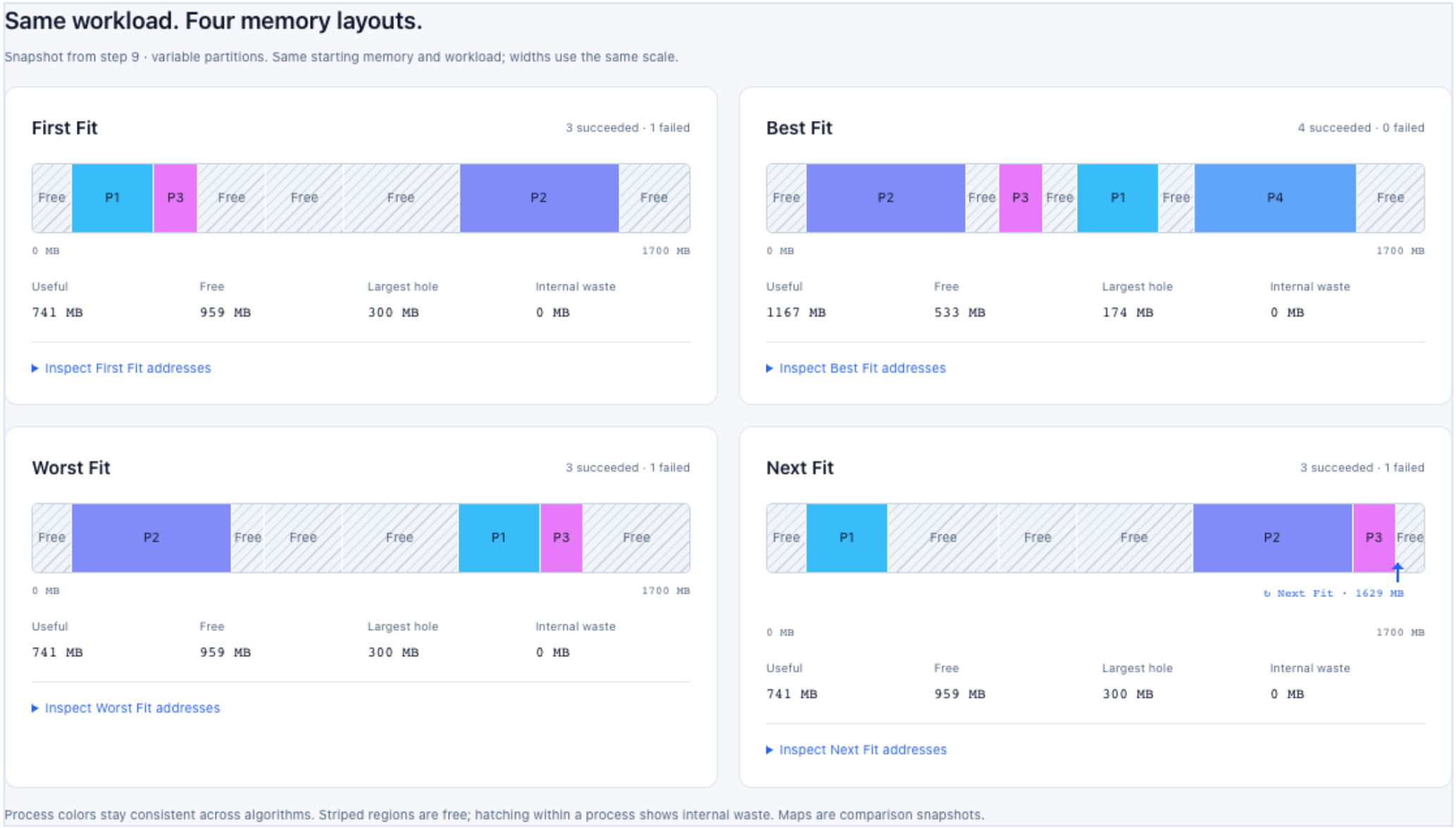
Mixed workload - First Fit / variable



Load Scenario sets up the requests. Next Action advances one lifecycle event. Run Workload completes the remainder.

Same workload, four different layouts

Classic variable-mode workload: 100, 500, 200, 300, 600 MB regions; requests 212, 417, 112, 426 MB.



Best Fit serves all four requests. First Fit, Worst Fit and Next Fit each serve three on this workload.

Compare the measured results

Exact website metrics: allocation success, useful utilization, fragmentation and search effort.

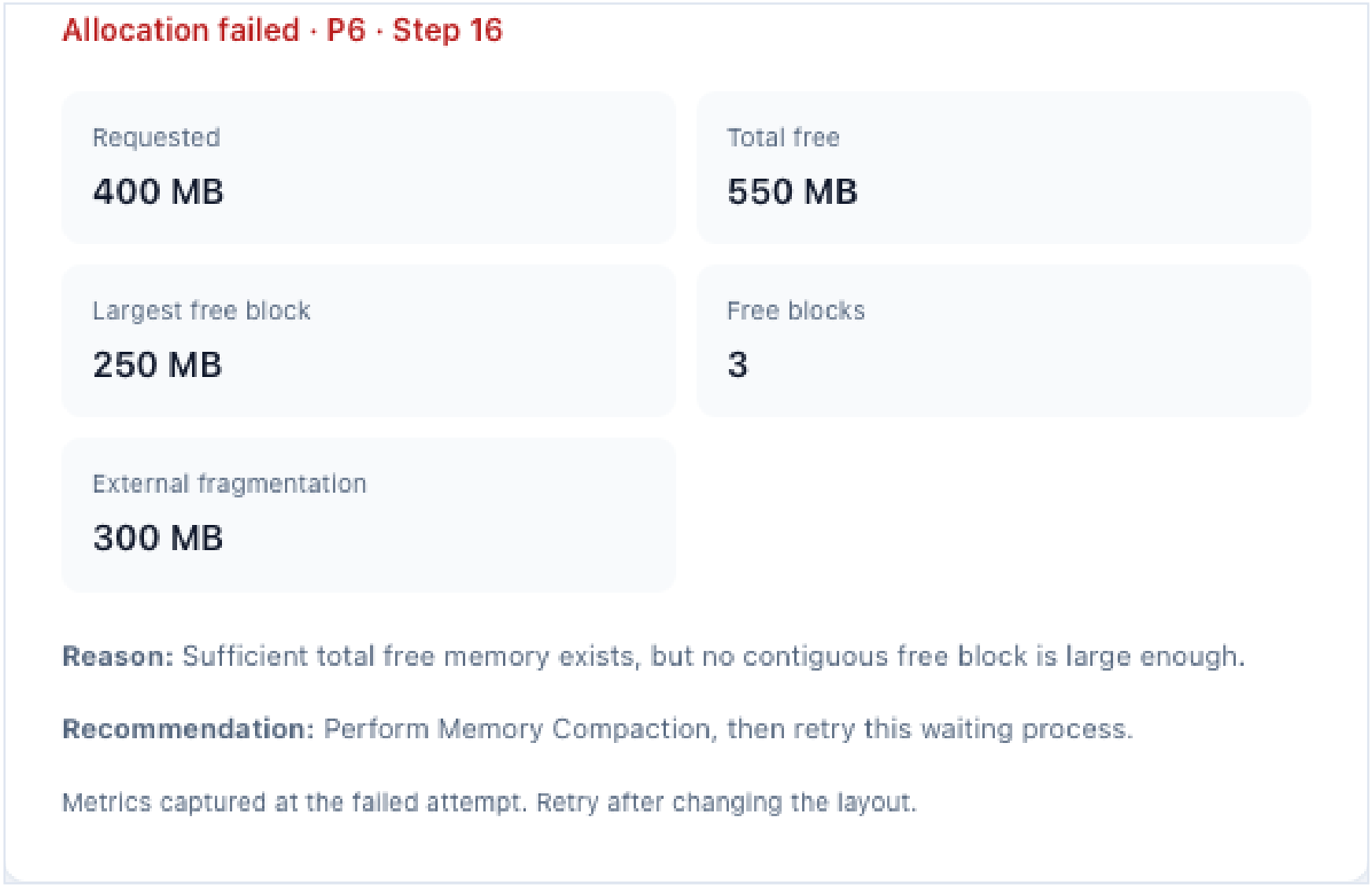
Metric	First Fit	Best Fit	Worst Fit	Next Fit
Successful allocations / growth	3	4	3	3
Failed allocations / growth	1	0	1	1
Success rate	75.00%	100.00%	75.00%	75.00%
Reserved memory utilization	43.59%	68.65%	43.59%	43.59%
Useful payload utilization	43.59%	68.65%	43.59%	43.59%
Final free memory	959 MB	533 MB	959 MB	959 MB
Largest free block	300 MB	174 MB	300 MB	300 MB
Free blocks	5	5	5	5
External fragmentation	659 MB	359 MB	659 MB	659 MB
External fragmentation ratio	68.72%	67.35%	68.72%	68.72%
Internal fragmentation	0 MB	0 MB	0 MB	0 MB
Internal fragmentation ratio	0.00%	0.00%	0.00%	0.00%
Search operations (free blocks)	14	20	20	12
Average search length	3.50	5.00	5.00	3.00
Efficiency score / 100	**61.48**	**80.19**	**59.98**	**61.98**

Best Fit: 4 successes, 0 failures, 68.65% useful utilization. The recommendation depends on this workload.

F56 | Successful/failed counts include growth where present. Search effort counts free regions examined.

Why enough free memory may still be unusable

Run the variable fragmentation workload and try the 400 MB request for P6.



550 MB is free, but the largest hole is only 250 MB. The 400 MB contiguous request therefore fails.

Compaction: before and after

Compaction moves allocated processes together and combines the separated free regions.

Step 17 · 2 processes moved

Before compaction



After compaction



Process	Before address	After address	Movement
P2	250–449	0–199	250 MB toward 0
P4	700–949	200–449	500 MB toward 0

Metric	Before	After
Free memory	550 MB	550 MB
Largest free block	250 MB	550 MB
External fragmentation	300 MB	0 MB

Snapshots capture the compaction itself; later actions appear in the live memory map. Free memory and process requests are preserved.

What changed?

P2: addresses 250-449 become 0-199.

P4: addresses 700-949 become 200-449.

Free memory stays 550 MB.

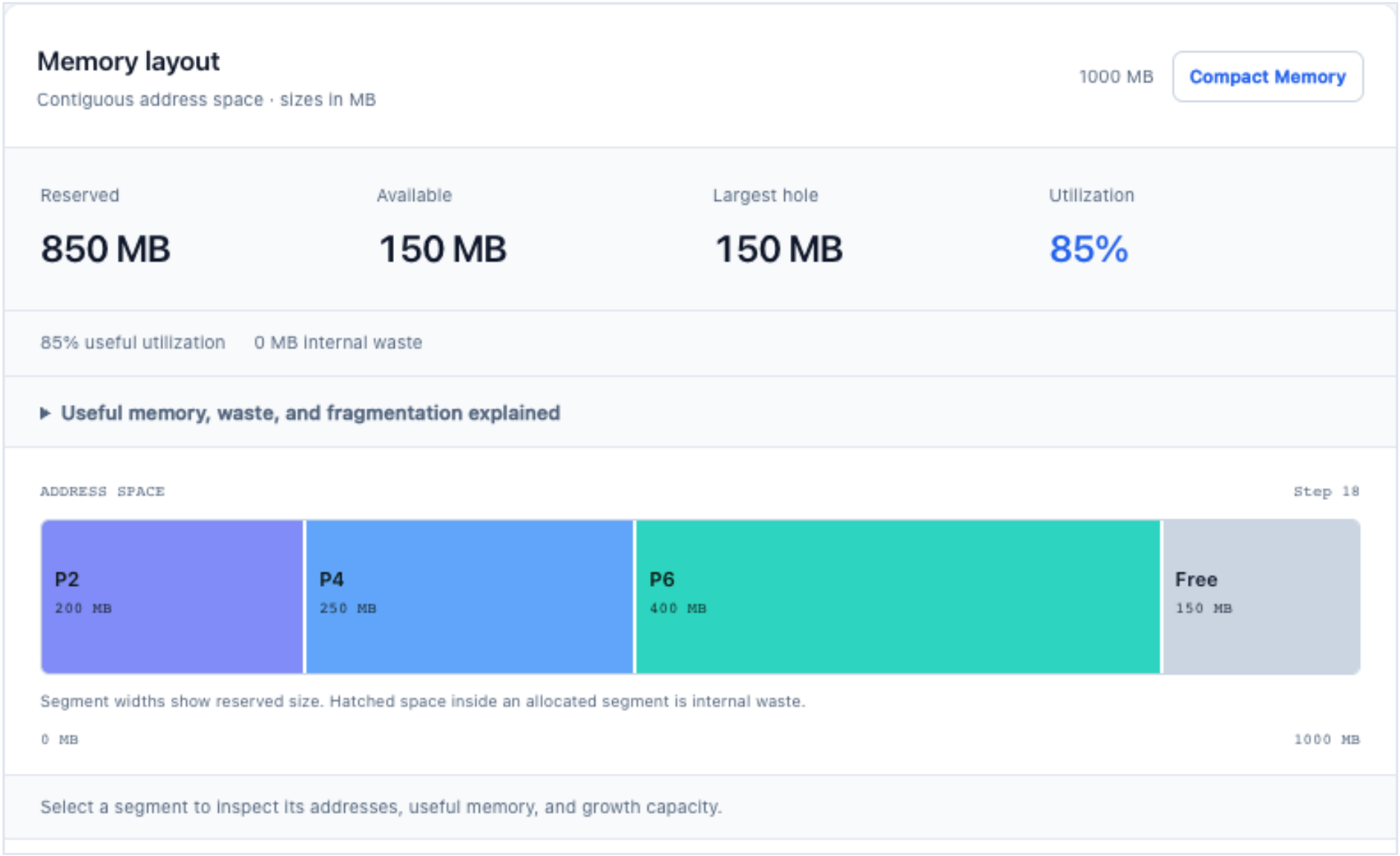
Largest hole grows from 250 MB to 550 MB.

External fragmentation falls from 300 MB to 0.

F11 | Native compaction capture: address movements and memory totals are shown together.

Retry succeeds after compaction

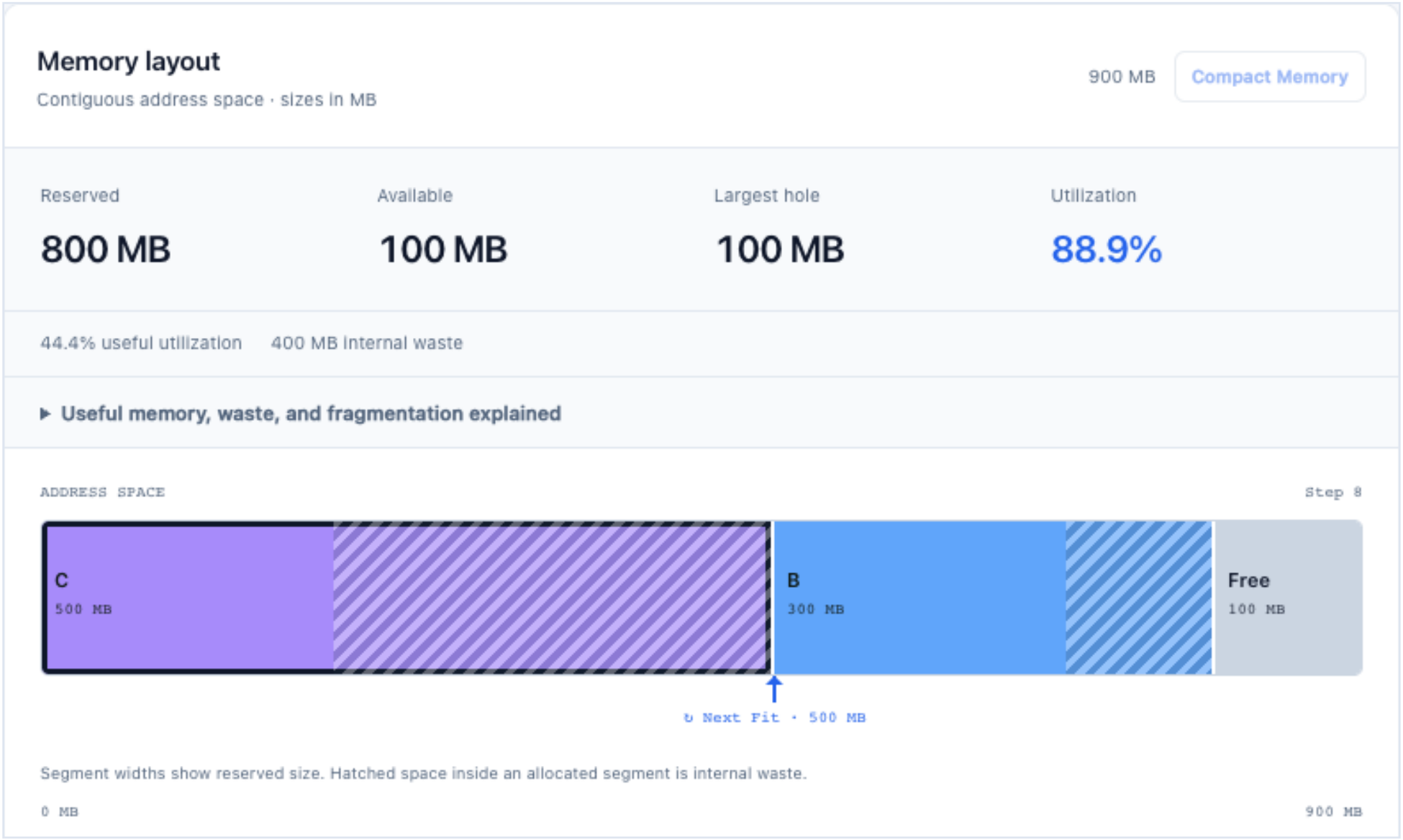
Retry the same P6 request after the free space becomes contiguous.



P6 receives 400 MB. Final state: 850 MB allocated, 150 MB free, 85% utilization.

Next Fit remembers where to search

Fixed partitions: 500, 300, 100 MB. Allocate A/B at 200 MB each; release A; allocate C at 200 MB.



Resume at 800 MB, skip the 100 MB tail, wrap to the released 500 MB partition; new cursor: 500 MB.

Fixed partitions reveal internal waste

A 200 MB process is allocated in a fixed 300 MB partition.

Memory layout

Contiguous address space · sizes in MB

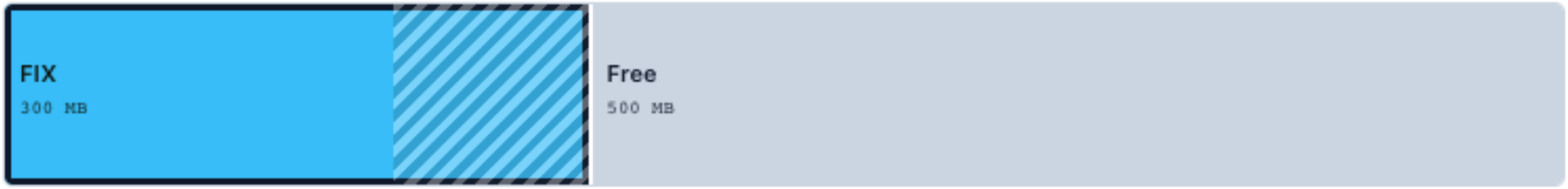
800 MB | Compact Memory

Reserved	Available	Largest hole	Utilization
300 MB	500 MB	500 MB	37.5%

25% useful utilization 100 MB internal waste

▶ Useful memory, waste, and fragmentation explained

ADDRESS SPACE Step 3



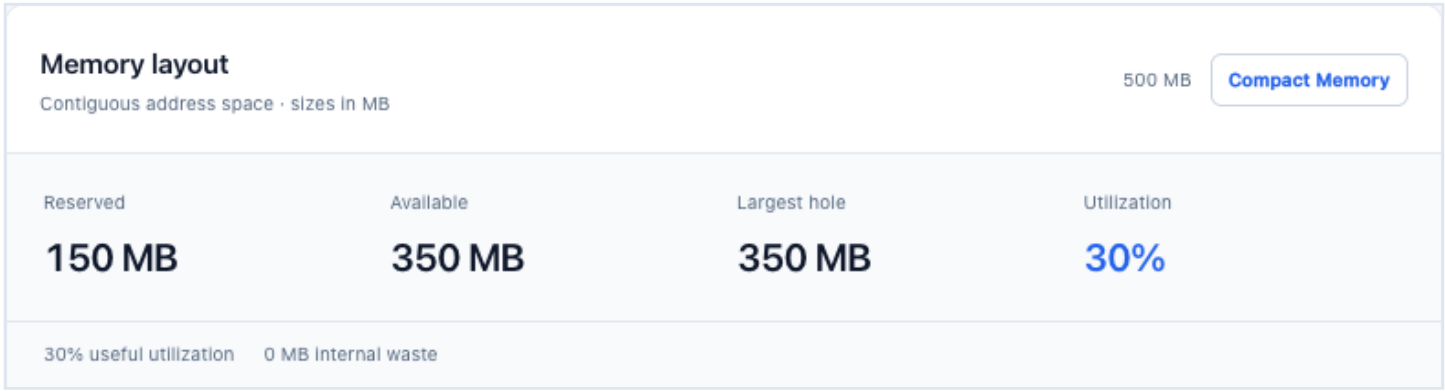
Segment widths show reserved size. Hatched space inside an allocated segment is internal waste.

0 MB 800 MB

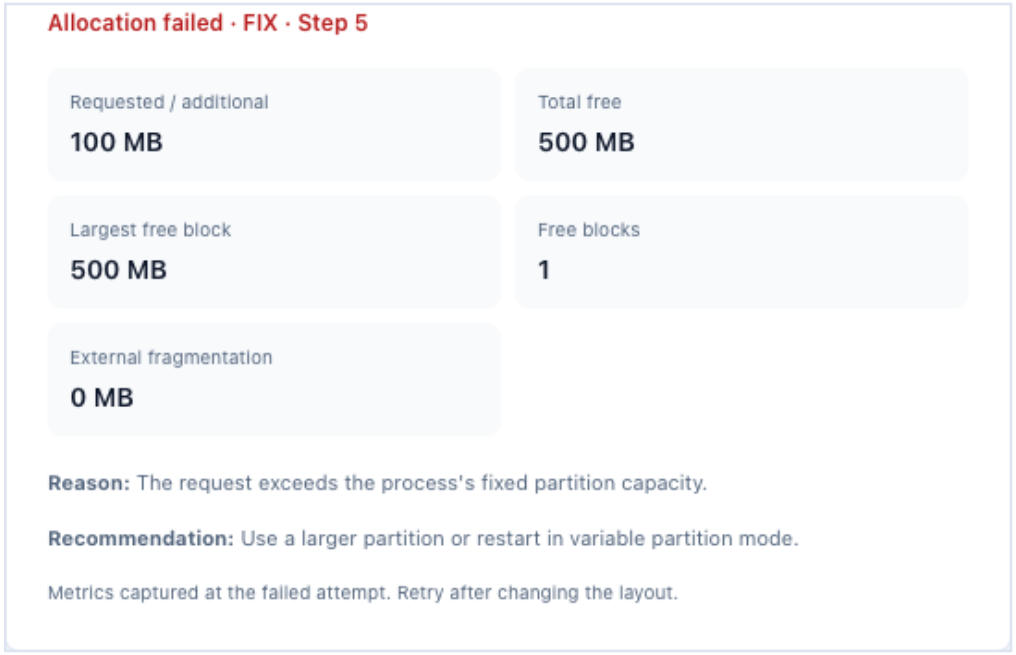
300 MB reserved - 200 MB useful = 100 MB internal fragmentation. Hatched space shows the waste.

Growth has limits in both partition modes

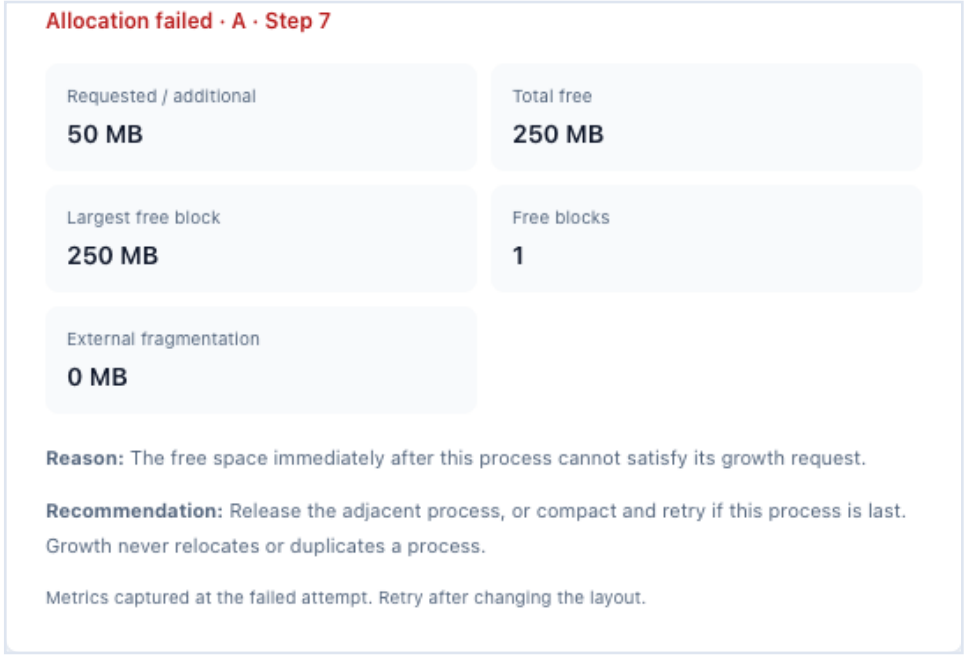
Existing allocations are preserved when an additional-memory request cannot fit in place.



Fixed: exceeds reserved partition



Variable: adjacent space is blocked

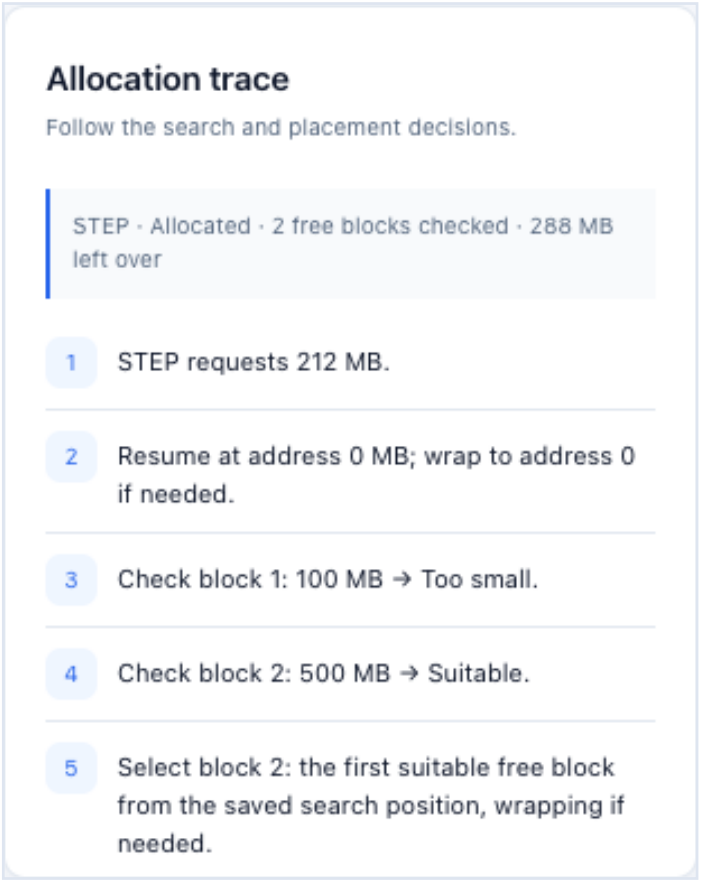


Top: A grows from 100 MB to 150 MB. Below: partition capacity and adjacency cause different failures.

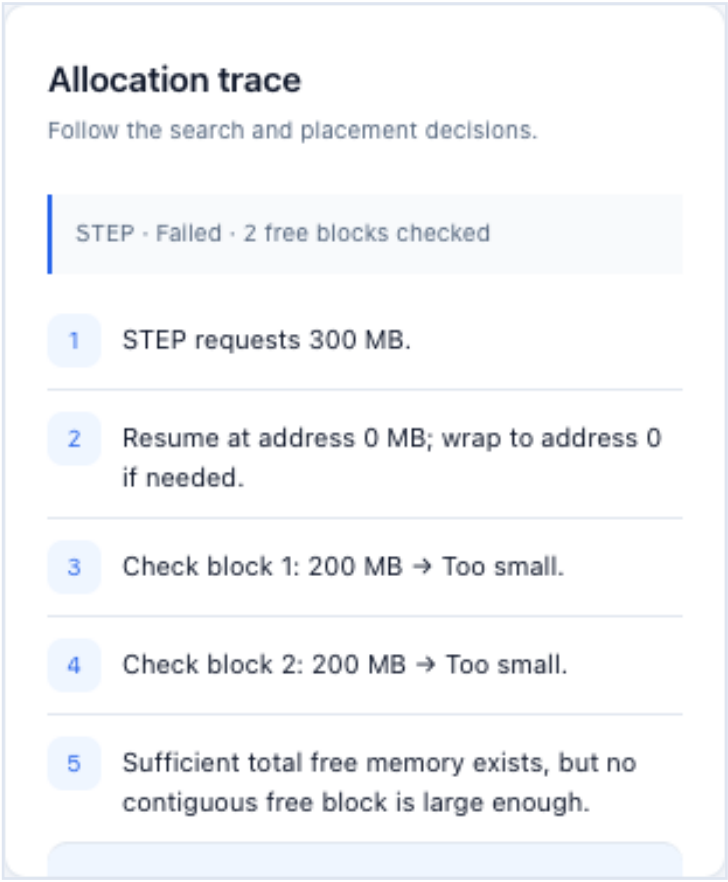
Watch the allocation decision step by step

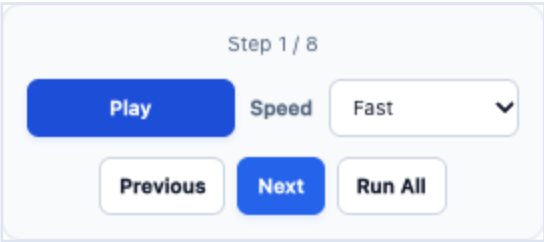
Next / Previous review the explanation; Run All or Play commits the result exactly once.

Successful Next Fit allocation



Failed Next Fit allocation

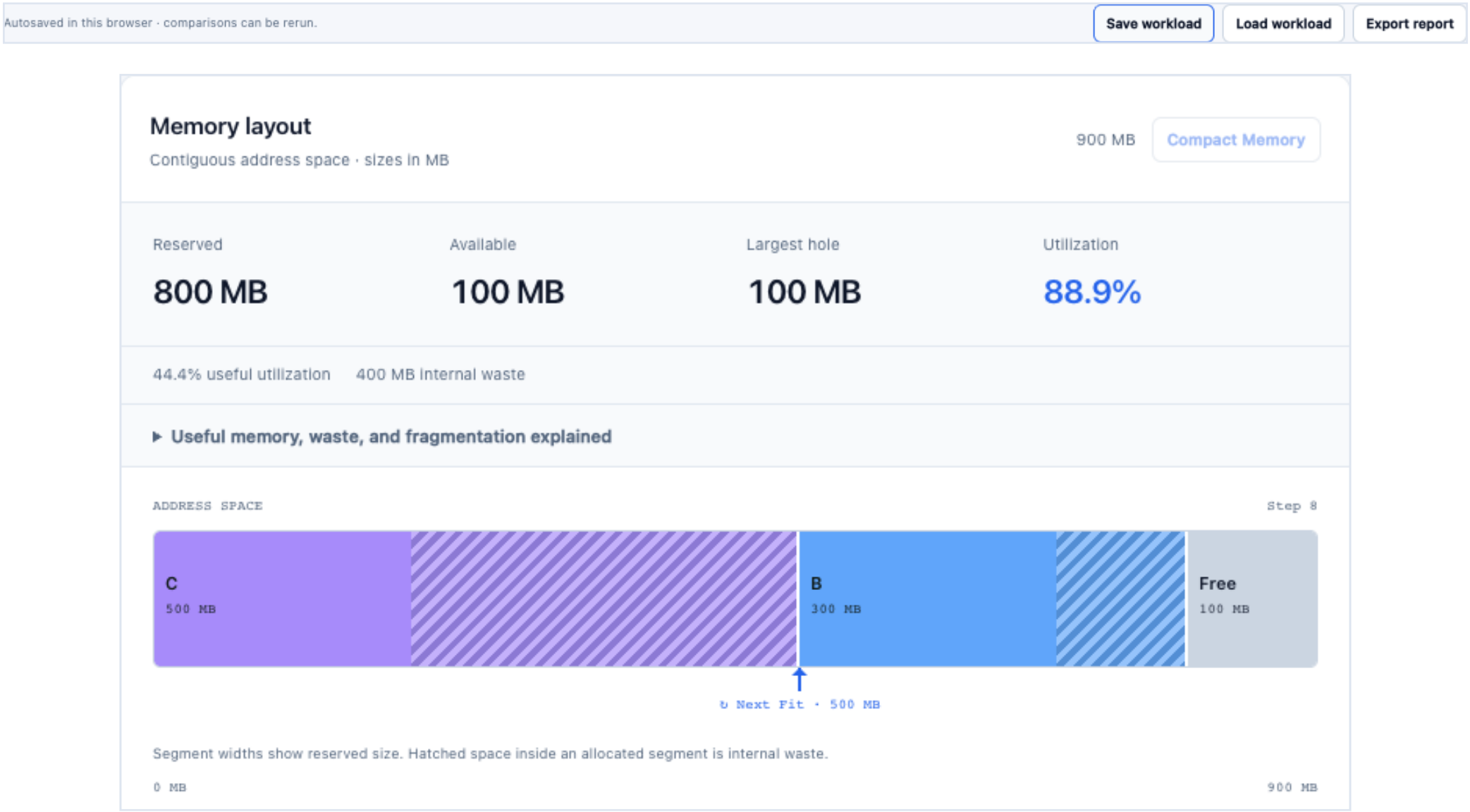




Play, Pause and speed selection are available. Success and failure walkthroughs passed for all four algorithms under both partition modes.

Save, load and resume your experiment

Download a portable workload, restore it through Load workload, or reload the browser autosave.

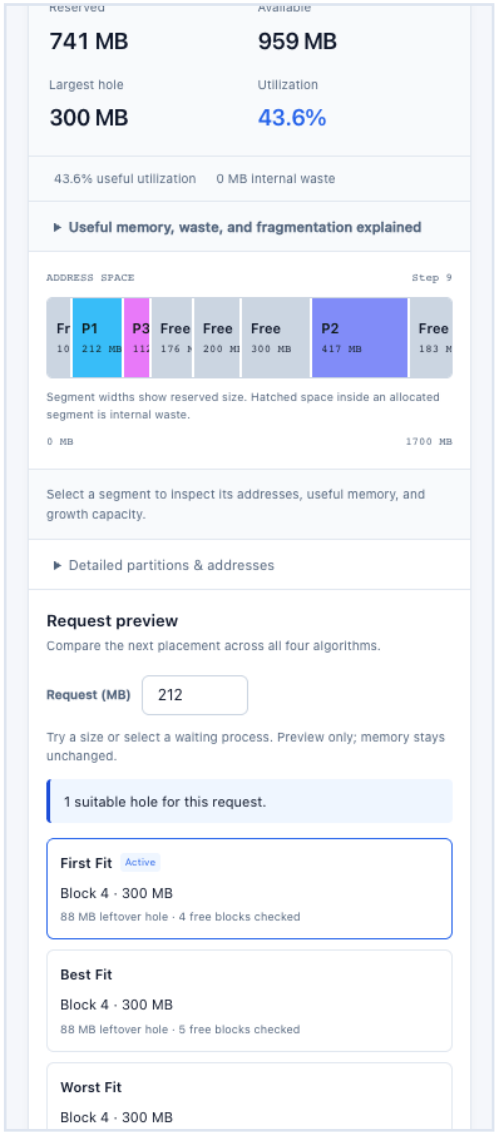


Reload reconstructs the same allocations and lifecycle history, including the Next Fit cursor at 500 MB.

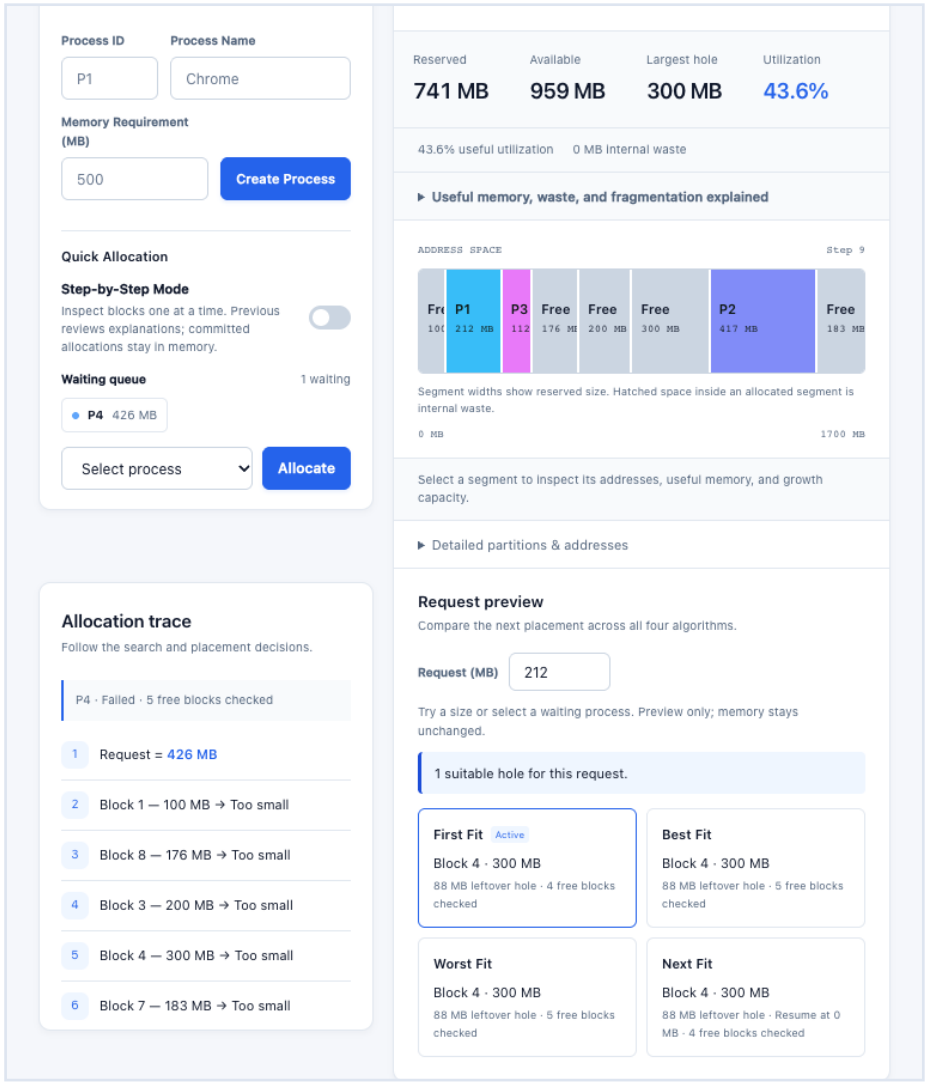
Use the website on smaller screens and in both themes

Responsive checks passed at widths 390, 768 and 1024 px; light and dark themes preserve the session.

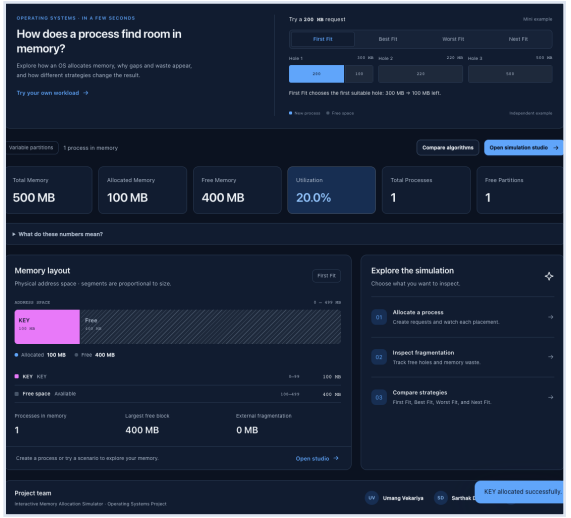
Phone: 390 px

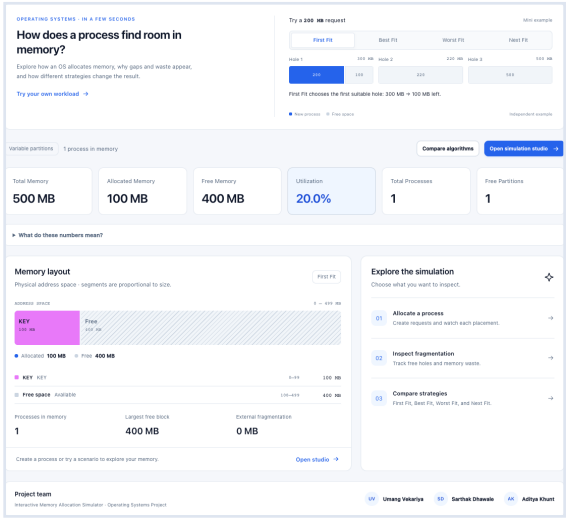


Tablet: 768 px



Themes





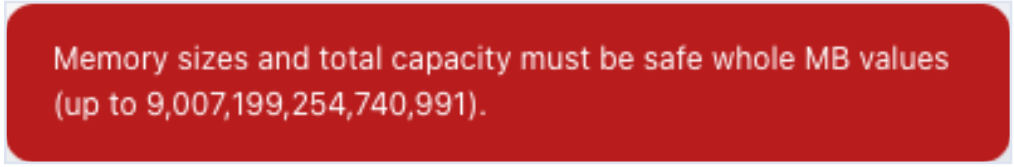
Full coverage summary and validation proof

All 40 built-in combinations passed, with 110 additional feature evidence checkpoints.

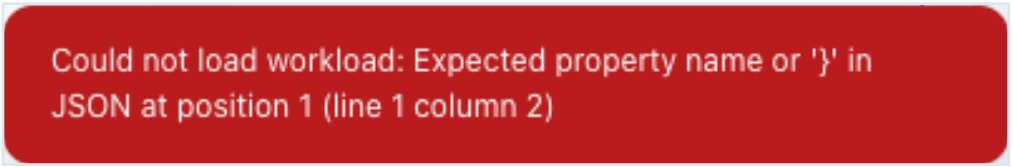
Workload	Var FF	Var BF	Var WF	Var NF	Fix FF	Fix BF	Fix WF	Fix NF
Normal	6 / 0	5 / 1	6 / 0	6 / 0	5 / 1	5 / 1	5 / 1	5 / 1
Fragmentation	5 / 1	5 / 1	5 / 1	5 / 1	2 / 4	2 / 4	2 / 4	2 / 4
Large	1 / 3	1 / 3	1 / 3	1 / 3	1 / 3	1 / 3	1 / 3	1 / 3
Mixed	5 / 2	5 / 2	6 / 1	5 / 2	5 / 2	5 / 2	5 / 2	5 / 2
Comparison	3 / 1	4 / 0	3 / 1	3 / 1	3 / 1	4 / 0	3 / 1	3 / 1

Cells = successful / failed allocation and growth attempts. FF/BF/WF/NF = First/Best/Worst/Next Fit.

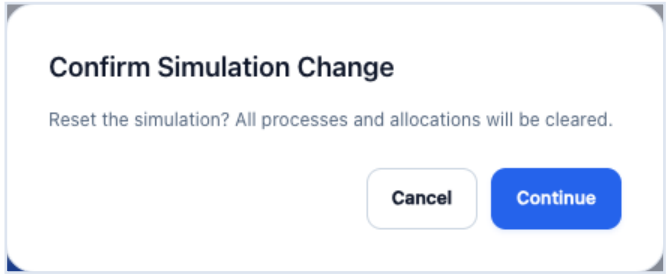
Unsafe memory values are rejected



Invalid workload import is rejected



Confirm before resetting the simulation



Tested in Chrome 155 on macOS. The existing regression suite also passed 48 checks, including 4,000 deterministic mixed actions.

This PDF shows representative captures. The separate archive retains all 150 screenshots, recorded states, coverage CSV and reusable workload/report downloads.